

4. The magnetic flux through a loop placed in a magnetic field can be changed by changing :
- (A) area of the loop only
 - (B) the value of magnetic field only
 - (C) orientation of the loop in the magnetic field only
 - (D) any one or more of the factors given in (A), (B) and (C)

CBSE 2026

4. The magnetic flux through a loop placed in a magnetic field can be changed by changing :
- (A) area of the loop only
 - (B) the value of magnetic field only
 - (C) orientation of the loop in the magnetic field only
 - (D) any one or more of the factors given in (A), (B) and (C)

CBSE 2026

(D) any one or more of the factors given in option (A), (B) and (C)

13. *Assertion (A) :* Induced emf produced in a coil will be more when the magnetic flux linked with the coil is more.

Reason (R) : Induced emf produced is directly proportional to the magnetic flux.

CBSE 2026

13. *Assertion (A) :* Induced emf produced in a coil will be more when the magnetic flux linked with the coil is more.

Reason (R) : Induced emf produced is directly proportional to the magnetic flux.

CBSE 2026

(D) Both Assertion(A) and reason (R) are false.

18. A plane circular coil is rotated about its vertical diameter with a constant angular speed ω in a uniform horizontal magnetic field. Initially the plane of the coil is parallel to the magnetic field. Draw plots showing the variation of the following physical quantities as the function of ωt , where t represents time elapsed :

CBSE 2026

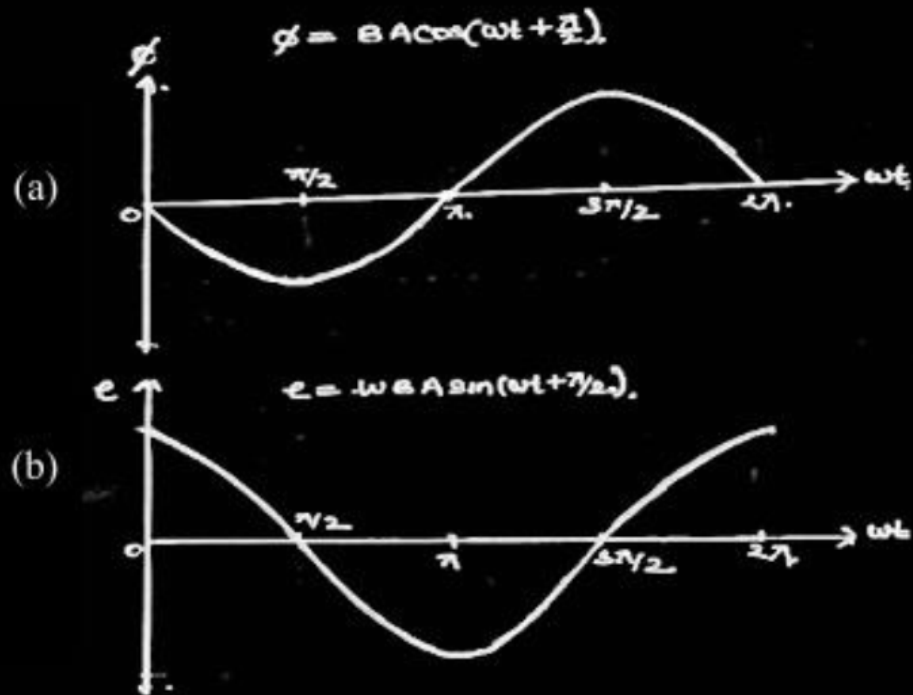
- (a) Magnetic flux ϕ linked with the coil, and
- (b) emf induced in the coil.

18. A plane circular coil is rotated about its vertical diameter with a constant angular speed ω in a uniform horizontal magnetic field. Initially the plane of the coil is parallel to the magnetic field. Draw plots showing the variation of the following physical quantities as the function of ωt , where t represents time elapsed :

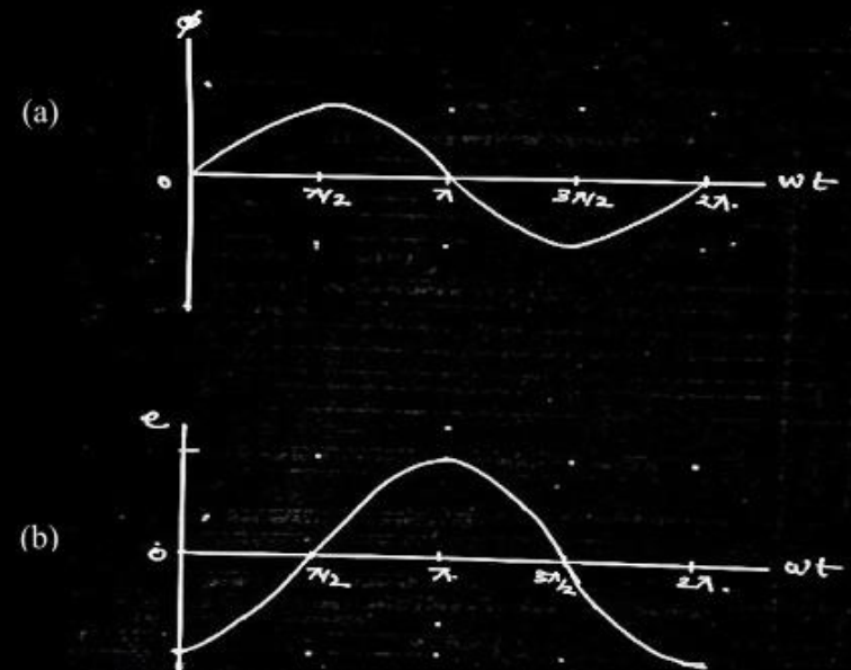
CBSE 2026

- Magnetic flux ϕ linked with the coil, and
- emf induced in the coil.

2



Alternatively,



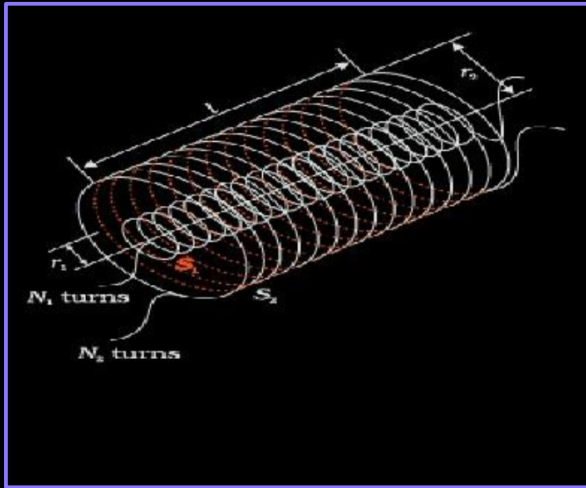
24. A long solenoid of length L and radius r_1 having N_1 turns is surrounded symmetrically by a coil of radius r_2 ($> r_1$) having N_2 turns ($N_2 \ll N_1$) around its mid-point. Derive an expression for the mutual inductance of solenoid and coil. Is $M_{12} = M_{21}$ valid in this case ?

CBSE 2026

24. A long solenoid of length L and radius r_1 having N_1 turns is surrounded symmetrically by a coil of radius r_2 ($> r_1$) having N_2 turns ($N_2 \ll N_1$) around its mid-point. Derive an expression for the mutual inductance of solenoid and coil. Is $M_{12} = M_{21}$ valid in this case ?

CBSE 2026

3



Let n_1 and n_2 be the total no. of turns per unit length of solenoid S_1 and coil S_2 respectively

$$N_1 = n_1 L, \quad N_2 = n_2 L$$

When current I_2 is set up in S_2 , flux linkage with S_1 is:

$$N_1 \phi_1 = N_1 A_1 B_2 \dots\dots\dots (i)$$

$\frac{1}{2}$

$$N_1 \phi_1 = (n_1 L)(\pi r_1^2)(\mu_o n_2 I_2)$$

$$N_1 \phi_1 = (\mu_o n_1 n_2 \pi r_1^2 L I_2) \dots\dots\dots (ii)$$

$\frac{1}{2}$

$$N_1 \phi_1 = M_{12} I_2 \dots\dots\dots (iii)$$

$\frac{1}{2}$

From (ii) and (iii)

$$M_{12} = (\mu_o n_1 n_2 \pi r_1^2 L)$$

$\frac{1}{2}$

Considering the reverse case, when I_1 current is set up in S_1 , flux linkage with S_2 is

$$N_2 \phi_2 = N_2 A_1 B_1 \dots\dots\dots (iv)$$

$$N_2 \phi_2 = (n_2 L)(\pi r_1^2)(\mu_o n_1 I_1) \dots\dots\dots (v)$$

$$N_2 \phi_2 = M_{21} I_1 \dots\dots\dots (vi)$$

From (v) and (vi)

$$M_{21} = (\mu_o n_1 n_2 \pi r_1^2 L)$$

$\frac{1}{2}$

Therefore, $M_{12} = M_{21}$

$\frac{1}{2}$

Alternatively,

$$M_{12} = M_{21} = \frac{(\mu_o N_1 N_2 \pi r_1^2)}{L}$$

9. A rectangular loop of size $5\text{ cm} \times 8\text{ cm}$ is lying in x-y plane in a uniform magnetic field $\vec{B} = (2.0\text{ T}) \hat{k}$. The total magnetic flux linked with the loop is :

(A) 80 Wb

(B) 16 Wb

(C) $8 \times 10^{-2}\text{ Wb}$

(D) $8 \times 10^{-3}\text{ Wb}$

CBSE 2026

9. A rectangular loop of size $5\text{ cm} \times 8\text{ cm}$ is lying in x-y plane in a uniform magnetic field $\vec{B} = (2.0\text{ T}) \hat{k}$. The total magnetic flux linked with the loop is :

(A) 80 Wb

(B) 16 Wb

(C) $8 \times 10^{-2}\text{ Wb}$

(D) $8 \times 10^{-3}\text{ Wb}$

CBSE 2026

(D) $8 \times 10^{-3}\text{ Wb}$

20. A square loop of side 10 cm, free to rotate about a vertical axis coinciding with its one arm, is initially held perpendicular to a uniform horizontal magnetic field of 0.2 T. If it is rotated at the uniform speed of 60 rpm, find the emf induced in the loop.

CBSE 2026

20. A square loop of side 10 cm, free to rotate about a vertical axis coinciding with its one arm, is initially held perpendicular to a uniform horizontal magnetic field of 0.2 T. If it is rotated at the uniform speed of 60 rpm, find the emf induced in the loop.

CBSE 2026

2

$$e = BA\omega \sin \omega t$$

$$e = 0.2 \times 100 \times 10^{-4} \times 2\pi \times \sin 2\pi t$$

$$e = 4\pi \times 10^{-3} \sin 2\pi t \text{ V}$$

$$= 12.56 \times 10^{-3} \sin 2\pi t \text{ V}$$

Note: Award full marks if $\sin 2\pi t$ is not mentioned in the final answer.

$\frac{1}{2}$

1

$\frac{1}{2}$

3. A square loop of side 0.50 m is placed in a uniform magnetic field of 0.4 T perpendicular to the plane of the loop. The loop is rotated through an angle of 60° in 0.2 s. The value of emf induced in the loop will be :

(A) 5 V

(B) 3.5 V

(C) 2.5 V

(D) Zero V

CBSE 2026

3. A square loop of side 0.50 m is placed in a uniform magnetic field of 0.4 T perpendicular to the plane of the loop. The loop is rotated through an angle of 60° in 0.2 s. The value of emf induced in the loop will be :

(A) 5 V

(B) 3.5 V

(C) 2.5 V

(D) Zero V

CBSE 2026

Award full mark for attempt

- 17.** A light copper ring is freely suspended by a light string. A bar magnet is held horizontally with its length along the axis of the ring. The magnet is moved towards the ring with its N pole facing the loop. What will happen to the ring and its position ? Explain.

CBSE 2026

17. A light copper ring is freely suspended by a light string. A bar magnet is held horizontally with its length along the axis of the ring. The magnet is moved towards the ring with its N pole facing the loop. What will happen to the ring and its position ? Explain.

CBSE 2026

2

Current is induced in the ring

$\frac{1}{2}$

Reason- Due to electromagnetic induction.

$\frac{1}{2}$

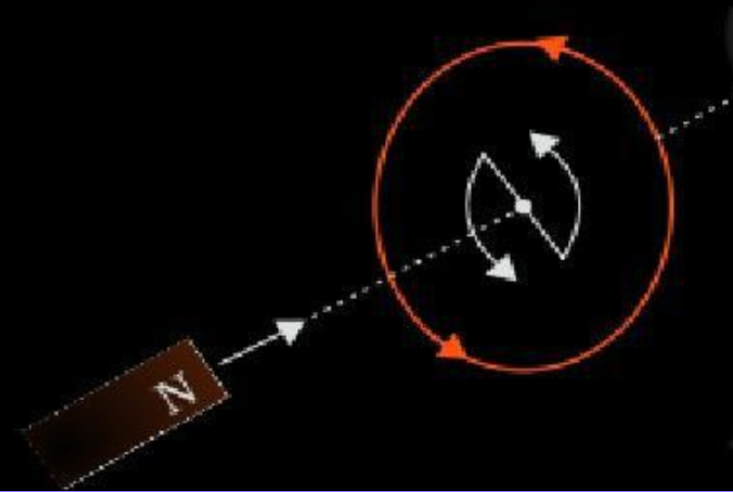
The ring will be repelled.

$\frac{1}{2}$

Reason- As per Lenz's law, Current is induced in the anticlockwise direction and magnetic moment associated with this current has north polarity.

$\frac{1}{2}$

Note: Award 1 mark if the student draws diagram showing current in anticlockwise direction.



4. The dimensions of the rate of change of magnetic flux are :
- (A) $[M L T^{-3} A]$ (B) $[M L^2 T^{-3} A^{-1}]$
(C) $[M L^2 T^{-2} A^{-1}]$ (D) $[M L^2 T^{-3} A^{-2}]$

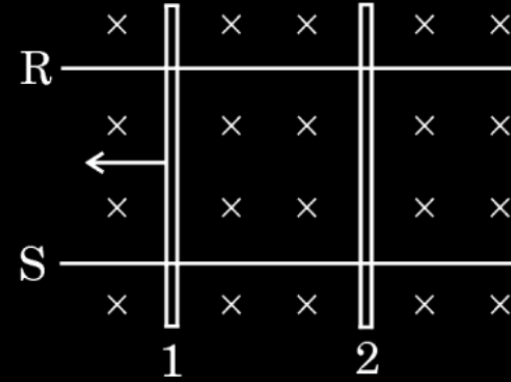
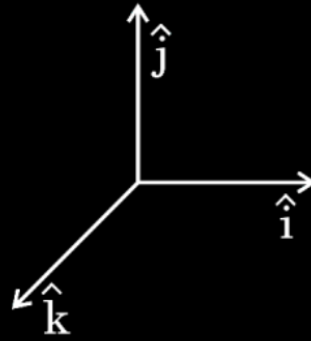
CBSE 2026

4. The dimensions of the rate of change of magnetic flux are :
- (A) $[M L T^{-3} A]$ (B) $[M L^2 T^{-3} A^{-1}]$
(C) $[M L^2 T^{-2} A^{-1}]$ (D) $[M L^2 T^{-3} A^{-2}]$

CBSE 2026

(B) $[ML^2T^{-3}A^{-1}]$

5. Two identical conductors 1 and 2 are placed on two frictionless conducting rails R and S in a uniform magnetic field directed vertically downward into the plane of the page. If conductor 1 is moved with a constant velocity in the direction as shown in figure, the force on conductor 2 will be along :



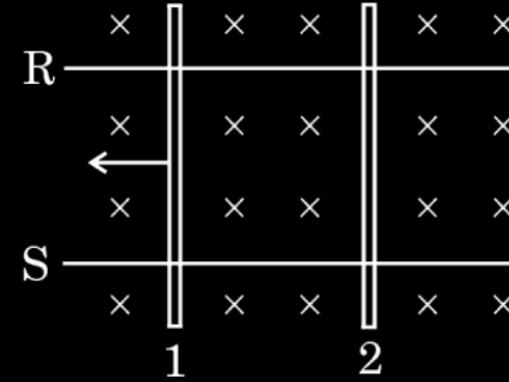
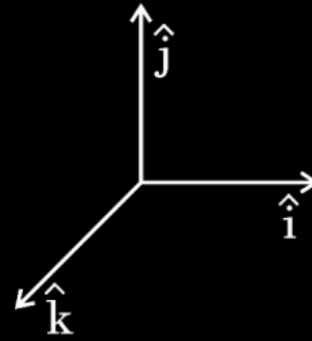
(A) $-\hat{i}$

(B) $-\hat{j}$

(C) $\hat{k}$

(D) $\hat{j}$

5. Two identical conductors 1 and 2 are placed on two frictionless conducting rails R and S in a uniform magnetic field directed vertically downward into the plane of the page. If conductor 1 is moved with a constant velocity in the direction as shown in figure, the force on conductor 2 will be along :



(A) $-\hat{i}$

(B) $-\hat{j}$

(C) $\hat{k}$

(D) $\hat{j}$

CBSE 2026

(A) $-\hat{i}$

(b) A rectangular loop of sides l and b and resistance 'R' is kept in a region in which the magnetic field varies as $B = B_0 \sin \omega t$.

CBSE 2026

(i) Derive expression for the emf induced in the loop.

(ii) Find the effective value of current that flows in the loop.

3

- (b) A rectangular loop of sides l and b and resistance 'R' is kept in a region in which the magnetic field varies as $B = B_0 \sin \omega t$.

CBSE 2026

- (i) Derive expression for the emf induced in the loop.
(ii) Find the effective value of current that flows in the loop.

3

(i) Deriving expression for induced emf	1 ½
(ii) Finding the effective value of current	1 ½

(i) Induced emf $= |\varepsilon| = \frac{d\phi}{dt}$
 $= \frac{d}{dt}(\vec{B} \cdot \vec{A})$
 $|\varepsilon| = \frac{d}{dt}(B_0 \sin \omega t \times \ell b)$

$$= B_0 \omega \ell b \cos \omega t$$

(ii) Instantaneous value of current $= \frac{|\varepsilon|}{R} = \frac{B_0 \omega \ell b \cos \omega t}{R}$

$$I_{eff} = \frac{I_0}{\sqrt{2}}$$

$$= \frac{B_0 \omega \ell b}{R\sqrt{2}}$$

½

½

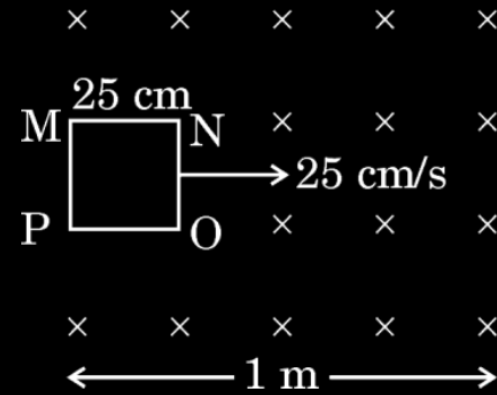
½

½

½

½

24. The figure given below shows a square-shaped loop MNOP of side 25 cm placed horizontally in a uniform magnetic field $\vec{B}$ directed vertically downward. The position of the loop at $t = 0$ s is as shown in figure. The loop is pulled with a constant velocity of 25 cm/s till it goes out of the field.



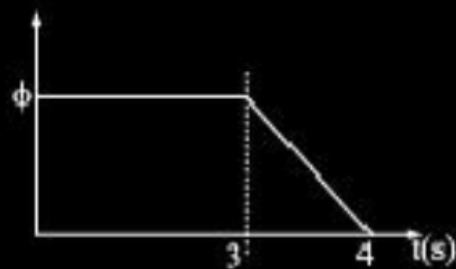
CBSE 2026

- (a) What will be the direction of the induced current in the loop as it goes out of the field ? For how long would the current in the loop persist ?
- (b) Plot graphs showing the variation of magnetic flux and magnitude of induced emf as a function of time.

(a) As the magnetic flux linked with the loop is decreasing, hence in accordance with Lenz's Law, the induced current will be in the clockwise direction.

$$t = \frac{25 \text{ cm}}{25 \text{ cm/s}} = 1 \text{ s}$$

(b)



1

1

$\frac{1}{2} + \frac{1}{2}$

2. A closely wound long solenoid of self-inductance L is cut into two identical solenoids. The value of self-inductance of each small solenoid will be :

(A) $\frac{L}{2}$

(B) $2L$

(C) $3L$

(D) $4L$

CBSE 2026

2. A closely wound long solenoid of self-inductance L is cut into two identical solenoids. The value of self-inductance of each small solenoid will be :

(A) $\frac{L}{2}$

(B) $2L$

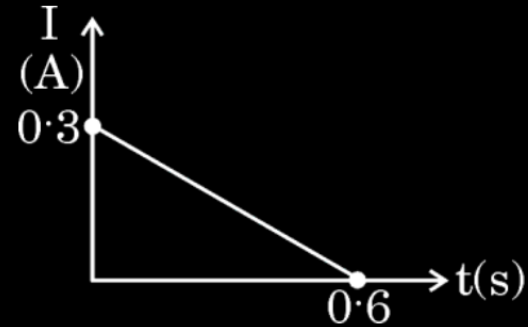
(C) $3L$

(D) $4L$

CBSE 2026

(A) $\frac{L}{2}$

27. A conducting rectangular loop of area 5 cm^2 and resistance 4Ω is removed from a region of uniform magnetic field, acting normal to the plane of the loop. The value of induced current I in the loop varies with time t , as shown in the figure.



Find :

- (a) total charge that passed through the loop
- (b) change in magnetic flux through the loop
- (c) magnitude of magnetic field in the region

CBSE 2026

(a) $Q = \text{Area under (I-t) curve}$ $= \frac{1}{2} \times \text{base} \times \text{height}$ $= \frac{1}{2} \times 0.6 \times 0.3 = 0.09 \text{ C}$	$\frac{1}{2}$
(b) Change in magnetic flux, $\Delta\phi = Q R$ $= 0.09 \times 4$ $= 0.36 \text{ Wb}$	$\frac{1}{2}$ $\frac{1}{2}$
(c) Magnitude of magnetic field $B = \frac{\Delta\phi}{A} = \frac{0.36}{5 \times 10^{-4}}$ $= 720 \text{ T}$	$\frac{1}{2}$ $\frac{1}{2}$

9. Two coils are placed closed to each other. The mutual inductance of the pair of coils depends upon the :
- (A) rate at which currents change in the two coils.
 - (B) relative position and orientation of the coils.
 - (C) currents in the two coils.
 - (D) value of voltage induced in one coil due to change in value of current in the other coil.

CBSE 2026

9. Two coils are placed closed to each other. The mutual inductance of the pair of coils depends upon the :
- (A) rate at which currents change in the two coils.
 - (B) relative position and orientation of the coils.
 - (C) currents in the two coils.
 - (D) value of voltage induced in one coil due to change in value of current in the other coil.

CBSE 2026

(B) relative position and orientation of the coils

- 28.** Two coils, one of radius 0.5 cm having 10 turns and the other of radius 5 cm having 50 turns are placed coaxially in air such that their centres are coincident. Calculate :
- (a) the magnetic flux through the smaller coil if the larger coil carries a current of 3 A, and
 - (b) the mutual inductance of the two coils.

28. Two coils, one of radius 0.5 cm having 10 turns and the other of radius 5 cm having 50 turns are placed coaxially in air such that their centres are coincident. Calculate :

- (a) the magnetic flux through the smaller coil if the larger coil carries a current of 3 A, and
- (b) the mutual inductance of the two coils.

3

CBSE 2026

$$(a) \phi_1 = N_1 B_2 A_1$$

1/2

$$= N_1 \times \frac{\mu_0 I_2}{2r_2} \times \pi r_1^2 \times N_2$$

1/2

$$= \frac{10 \times 4\pi \times 10^{-7} \times 3 \times 3.14 \times 0.5 \times 0.5 \times 10^{-4} \times 50}{2 \times 5 \times 10^{-2}}$$

1/2

$$= 14.789 \times 10^{-7} \text{ Wb}$$

$$(b) M_{12} = \frac{\phi_1}{I_2}$$

1/2

$$= \frac{14.789 \times 10^{-7}}{3}$$

1/2

$$= 4.92 \times 10^{-7} \text{ H}$$

1/2

2. A square loop of side 50 cm is placed in a uniform magnetic field of 3.0 T acting perpendicular to the plane of the loop. If the loop is rotated through an angle of 90° in 0.3 s, the value of emf induced in the loop would be :

- (A) 0.25 V
- (B) 0.50 V
- (C) 0.75 V
- (D) 1.0 V

CBSE 2026

2. A square loop of side 50 cm is placed in a uniform magnetic field of 3.0 T acting perpendicular to the plane of the loop. If the loop is rotated through an angle of 90° in 0.3 s, the value of emf induced in the loop would be :

- (A) 0.25 V
- (B) 0.50 V
- (C) 0.75 V
- (D) 1.0 V

CBSE 2026

No option is correct (Award full credit for attempting)

- (b) (i) State Lenz's law and explain that it follows the law of conservation of energy.
- (ii) Write the dimensional formula for self-inductance.

The current in a coil changes from 8.0 A to 2.0 A in 0.6 s . If an average emf induced in the coil is 50 V , calculate the self-inductance of the coil.

CBSE 2026

(i)	The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produces it. Consider north pole of a magnet being moved towards a coil. The magnetic flux through the coil increases, hence current is induced in counter clockwise direction making the side of the coil behave as north pole causing repulsion. If this is not the case then bringing magnet towards a coil will construct a perpetual motion machine by a suitable arrangement. Therefore production of induced emf in a direction which opposes the change in magnetic flux follows from law of conservation of energy.	1
(ii)	$[ML^2T^{-2}A^{-2}]$ $\varepsilon = -L \frac{dI}{dt}$ $L = \frac{-(50) \times 0.6}{(2 - 8)}$ $= 5 \text{ H}$	1
		1
		$\frac{1}{2}$
		1
		$\frac{1}{2}$

7. A metallic rod of 1 m length is rotated with a frequency of 40 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field of 0.5 T parallel to the axis is present in the region. The value of emf induced between the centre and the metallic ring is close to :

(A) 20 V

(B) 32 V

(C) 40 V

(D) 63 V

CBSE 2026

7. A metallic rod of 1 m length is rotated with a frequency of 40 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field of 0.5 T parallel to the axis is present in the region. The value of emf induced between the centre and the metallic ring is close to :

(A) 20 V

(B) 32 V

(C) 40 V

(D) 63 V

CBSE 2026

(D) 63 V

4. A square loop of side L lies in the x - y plane in a magnetic field $\vec{B} = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})$, where B_0 is a constant. The magnetic flux through the loop is :

(A) $\sqrt{29} B_0 L^2$

(B) $4 B_0 L^2$

(C) $3 B_0 L^2$

(D) $2 B_0 L^2$

CBSE 2026

4. A square loop of side L lies in the x - y plane in a magnetic field $\vec{B} = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})$, where B_0 is a constant. The magnetic flux through the loop is :

(A) $\sqrt{29} B_0 L^2$

(B) $4 B_0 L^2$

(C) $3 B_0 L^2$

(D) $2 B_0 L^2$

CBSE 2026

(B) $4 B_0 L^2$

3. An electric current can be induced in a wire loop if –

1

- (A) the wire loop is wound around a bar magnet.
- (B) a bar magnet is placed in contact with the wire loop.
- (C) a bar magnet is placed near the wire loop.
- (D) a bar magnet is moved rapidly towards the wire loop.

CBSE 2026

3. An electric current can be induced in a wire loop if –

1

- (A) the wire loop is wound around a bar magnet.
- (B) a bar magnet is placed in contact with the wire loop.
- (C) a bar magnet is placed near the wire loop.
- (D) a bar magnet is moved rapidly towards the wire loop.

CBSE 2026

(D) a bar magnet is moved rapidly toward the wire loop

24. (a) Briefly explain how Lenz's law is consistent with the law of conservation of energy.
- (b) Considering the case of the magnetic field B inside a solenoid of length l , area of cross-section A , prove that magnetic energy per unit volume is $B^2/(2 \mu_0)$.

CBSE 2026

3

24. (a) Briefly explain how Lenz's law is consistent with the law of conservation of energy.
- (b) Considering the case of the magnetic field B inside a solenoid of length l , area of cross-section A , prove that magnetic energy per unit volume is $B^2/(2\mu_0)$.

CBSE 2026

3

(a) Lenz's law is consistent with the law of conservation of energy because:
When a magnet is moved towards or away from a coil, then induced current creates its own magnetic field that opposes the motion of the magnet. Due to this opposition an external force must be applied to keep the magnet moving. The work done by this external force is converted into electrical energy.

1

(b) Magnetic field inside a solenoid, $B = \mu_0 n I$

$\frac{1}{2}$

Coefficient of self-inductance, $L = \mu_0 n^2 A l$

$\frac{1}{2}$

Magnetic energy $U_B = \frac{1}{2} L I^2$

$\frac{1}{2}$

$$= \frac{1}{2} (\mu_0 n^2 A l) \left(\frac{B}{\mu_0 n} \right)^2$$

$$= \frac{1}{2\mu_0} B^2 A l$$

Magnetic energy per unit volume, $\frac{U_B}{Volume} = \frac{U_B}{A l} = \frac{B^2}{2\mu_0}$

$\frac{1}{2}$

32. (a) Define mutual inductance. Give its units. State two factors on which the mutual inductance between a given pair of coils depends.
- (b) A 1.0 m long solenoid of radius 2 cm having 2000 turns has a secondary coil of 1000 turns wound closely near its mid-point. Find the mutual inductance between the two coils.

5

CBSE 2026

(i) Mutual inductance is defined as the induced emf in primary coil when the current in secondary coil changes at the unit rate. Alternatively, Mutual inductance is defined as the magnetic flux linked with the primary coil when the current in secondary coil is unity. S.I unit: henry (H) Alternatively, weber/ampere Alternatively, Tm^2A^{-1} Factors: (1) Separation between two coils (2) Relative orientation between two coils (3) Nature of the medium (Any two)	1 1 $\frac{1}{2}$ $\frac{1}{2}$ 1
(ii) Mutual Inductance, $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$ $= \frac{(4\pi \times 10^{-7})(2000)(1000)(3.14)(2 \times 10^{-2})^2}{1}$ $= 3.16 \times 10^{-3} \text{H}$	$\frac{1}{2}$ $\frac{1}{2}$

8. The magnetic flux ϕ (in Wb) linked with a coil is related to time t (in s) as
 $\phi = 5 At^2 + Bt - 2C$

The SI units of A and B are respectively

1

(A) $\text{Wb s}^2, \text{Wb s}$

(B) $\text{Wb s}^{-1}, \text{Wb}$

(C) $\text{Wb s}^{-2}, \text{Wb s}^{-1}$

(D) $\text{Wb s}^{-1}, \text{Wb s}^{-2}$

CBSE 2026

8. The magnetic flux ϕ (in Wb) linked with a coil is related to time t (in s) as
$$\phi = 5 At^2 + Bt - 2C$$

The SI units of A and B are respectively

1

(A) $\text{Wb s}^2, \text{Wb s}$

(B) $\text{Wb s}^{-1}, \text{Wb}$

(C) $\text{Wb s}^{-2}, \text{Wb s}^{-1}$

(D) $\text{Wb s}^{-1}, \text{Wb s}^{-2}$

CBSE 2026

(C) $\text{Wb s}^{-2}, \text{Wb s}^{-1}$

10. A magnet held vertically, with its north pole down is dropped along the axis of a closed solenoid placed vertically on a table. If the observer looks down from the top,

1

- (A) the induced current will flow in the anticlockwise direction.
- (B) the induced current will flow in the clockwise direction.
- (C) no induced current will flow in the solenoid.
- (D) the magnet will fall with a constant velocity.

CBSE 2026

10. A magnet held vertically, with its north pole down is dropped along the axis of a closed solenoid placed vertically on a table. If the observer looks down from the top,

1

- (A) the induced current will flow in the anticlockwise direction.
- (B) the induced current will flow in the clockwise direction.
- (C) no induced current will flow in the solenoid.
- (D) the magnet will fall with a constant velocity.

CBSE 2026

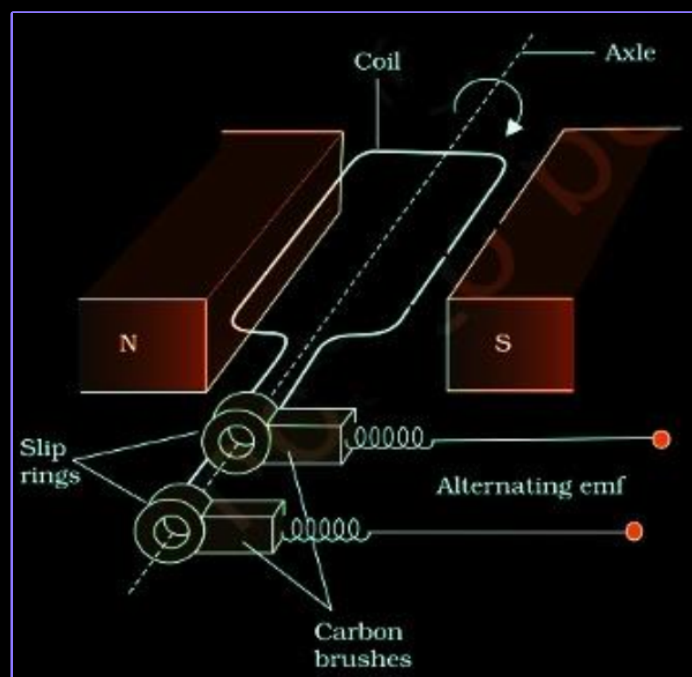
(A) The induced current will flow in the anticlockwise direction

25. (i) Define mutual inductance of a pair of coils. Write its SI unit. 3
- (ii) A long solenoid of radius R and length L has n turns per unit length. A circular loop of radius $r (< R)$ is placed inside at the centre of the solenoid such that its axis coincides with the axis of the solenoid. Obtain the mutual inductance of the solenoid and the loop. CBSE 2026

(i)	It is the ratio of the magnetic flux linked with a coil to the current flowing in the neighbouring coil. <u>Alternatively:</u> It is the ratio of magnitude of induced emf in a coil to the rate of change of current in the neighbouring coil. Note: Award ($1/2$) mark if a student writes $M = \frac{\phi}{I}$ OR $M = \frac{ \varepsilon }{di/dt}$ SI unit: henry(H) Or Tm^2A^{-1} Or WbA^{-1} or VsA^{-1}	1
(ii)	If current I is setup through the solenoid, the magnetic field inside at its center $B = \mu_0 n I$ Flux linked with the circular loop of radius $r(< R)$ $\phi_B = BA$ $\phi_B = (\mu_0 n I)(\pi r^2)$ Mutual Inductance $M = \frac{\phi_B}{I}$ $M = \mu_0 n \pi r^2$	$1/2$ $1/2$ $1/2$ $1/2$

- (i) With the help of a labelled diagram, explain the principle, construction and working of an a.c. generator.
- (ii) Deduce an expression for the induced emf in the coil of the generator.
- (iii) If T is the time period of the rotation of the coil, at what values of T in a cycle, the emf generator is maximum ?

CBSE 2026



• Principle: It is based on the principle of electromagnetic induction. When the magnetic flux passing through a coil changes continuously, emf is induced.

Working: When coil is rotated in magnetic field, magnetic flux linked with it changes and emf is induced in the coil.

(ii) Flux linked with a coil at any instant of time placed in magnetic field rotating with angular speed ω

$$\phi_B = BA \cos \omega t$$

From faraday's law

$$\varepsilon = -N \frac{d\phi_B}{dt}$$

$$\varepsilon = NBA\omega \sin \omega t$$

(iii) EMF generated is maximum at $\frac{T}{4}$ and $\frac{3T}{4}$

1

1

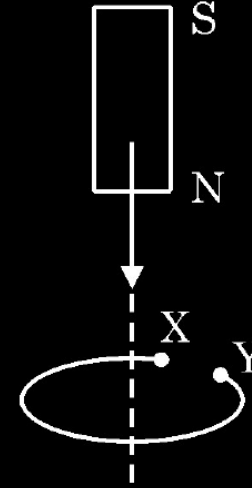
1/2

1/2

1

6. Figure shows a magnet dropped through a small loop with a small cut. Which of the following statements is correct ?

1

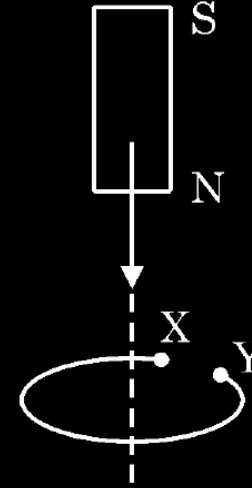


CBSE 2026

- (A) The speed of the falling magnet increases as it approaches the loop and starts decreasing as it crosses the loop.
- (B) Acceleration of magnet increases as it approaches the loop and starts decreasing as it crosses the loop.
- (C) Speed of magnet remains uniform as it moves through the loop.
- (D) Acceleration of the magnet remains uniform as it moves through the loop.

6. Figure shows a magnet dropped through a small loop with a small cut. Which of the following statements is correct ?

1



CBSE 2026

- (A) The speed of the falling magnet increases as it approaches the loop and starts decreasing as it crosses the loop.
- (B) Acceleration of magnet increases as it approaches the loop and starts decreasing as it crosses the loop.
- (C) Speed of magnet remains uniform as it moves through the loop.
- (D) Acceleration of the magnet remains uniform as it moves through the loop.

(D) Acceleration of the magnet remains uniform as it moves through the loop.

8. Two identical circular loops A and B of metal wire are arranged on a table close to each other. Loop A is connected to a battery and the current in it increases with time. In response, the loop B

1

- (A) remains stationary
- (B) is attracted by loop A
- (C) is repelled by loop A
- (D) rotates about its centre of mass which is fixed

CBSE 2026

8. Two identical circular loops A and B of metal wire are arranged on a table close to each other. Loop A is connected to a battery and the current in it increases with time. In response, the loop B

1

- (A) remains stationary
- (B) is attracted by loop A
- (C) is repelled by loop A
- (D) rotates about its centre of mass which is fixed

CBSE 2026

(C) is repelled by loop A

33. (a) State Faraday's law of electromagnetic induction.

5

(b) Derive an expression for the self-inductance of an air-filled long solenoid of length l and cross-sectional area A having N turns.

(c) A conducting rod of length 50 cm, with one end pivoted, is rotated with angular speed of 60 rpm in a uniform magnetic field of 4.0 mT directed perpendicular to the plane of rotation of rod. Find the emf induced in the rod.

CBSE 2026

(a) The magnitude of the induced e.m.f in a circuit is equal to the time rate of change of magnetic flux through the circuit.	1
Alternatively:	
$\varepsilon = -\frac{d\phi_B}{dt}$	
(b) Magnetic field due to current carrying long solenoid of length l area of cross section A having n turns per unit length is $B = \mu_0 nI$	
Total magnetic flux linked with the solenoid is	$\frac{1}{2}$
$N\phi_B = (nl)(\mu_0 nI)(A)$	
$= \mu_0 n^2 A l I$	$\frac{1}{2}$
Where nl is total number of turns. Thus	
$\therefore$ Self inductance of the solenoid	
$L = \frac{N\phi_B}{I}$	$\frac{1}{2}$
$L = \mu_0 n^2 A l$	
Note: Award full marks for any other correct alternative method	$\frac{1}{2}$
(c) Induced e.m.f	
$\varepsilon = \frac{1}{2} B l^2 \omega$	$\frac{1}{2}$
$= \frac{1}{2} \times 4 \times 10^{-3} \times (50 \times 10^{-2})^2 \times (2\pi \times 1)$	1
$= 3.14 \text{ mV}$	$\frac{1}{2}$